# Autonomous Trading Agent using Deep Reinforcement Learning

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**Abstract**

This project explores the application of Deep Reinforcement Learning (DRL) for autonomous financial trading by implementing three distinct algorithms, Advantage Actor-Critic (A2C), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO), to actively trade the S&P 500 (SPY). To accurately capture market dynamics, we designed a custom trading environment using established trading fundamentals and technical indicators, including the Volatility Index (VIX), Relative Strength Index (RSI), MACD, etc, along with our own unique reward function. After training the agents and evaluating them on an out-of-sample period, all three models successfully beat the traditional Buy & Hold baseline strategy. Ultimately, PPO emerged as the best-performing model, demonstrating superior robustness against market volatility and achieving the highest overall Sharpe Ratio of 1.37.

## 1 Introduction

Financial market trading is a notoriously complex task characterized by high volatility and non-stationary data. Traditional trading strategies often rely on rigid rules or linear models that struggle to adapt to these rapidly changing market conditions. To overcome these limitations, Deep Reinforcement Learning (DRL) has emerged as a powerful alternative. Instead of merely predicting future prices, a DRL agent learns optimal, adaptive trading policies directly through continuous trial-and-error interactions with the market environment. This project leverages the adaptive capabilities of DRL to create an autonomous trading agent for the S&P 500 index (SPY).

## 2 Related Work

Recent literature has increasingly explored Deep Reinforcement Learning (DRL) to navigate the stochastic nature of financial markets. Yang et al. [3] proposed an automated stock trading system utilizing an ensemble of three actor-critic algorithms: Advantage Actor-Critic (A2C), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO). By aggregating the outputs of these models, their ensemble strategy robustly adapted to varying market conditions, outperforming both the Dow Jones Industrial Average (DJIA) and traditional min-variance portfolios in risk-adjusted returns.

Similarly, a broader comparative study on actor-critic methods [2] evaluated five DRL algorithms as stock trading controllers, which included A2C, DDPG, PPO, TD3, and SAC. The study concluded that all five algorithms successfully beat the DJIA baseline, with PPO achieving the highest overall cumulative returns. Finally, practical implementations of PPO for dynamic portfolio management have demonstrated the algorithm's specific efficacy in processing historical market data and technical indicators to autonomously optimize discrete trading actions (Buy, Hold, Sell) over time [1]. With these works, we see a common pattern of the superiority of actor-critic architectures in terms of performance in financial markets, especially PPO. Therefore, in our study, we will be incorporating some of these models in our methodology, and monitor their effectiveness compared to standard trading strategies such as the buy and hold.

# 3 Problem Statement and Objective

The primary objective of this project is to develop an autonomous trading system for the S&P 500 index (SPY) capable of generating profits with superior risk-adjusted returns compared to a standard market baseline, which is the Buy and Hold Strategy.

# 4 Methodology

## 4.1 Data

We used historical daily data for the S&P 500 ETF (SPY) and the Volatility Index (VIX) sourced from Yahoo Finance. The dataset is explicitly partitioned into a 10-year training period (2015-2024) and a 1-year out-of-sample testing period (2025). To align our agents with real-world professional trading practices, the state space was enriched with standard technical indicators: MACD (measuring trend direction and momentum), 14-day RSI (identifying overbought or oversold market conditions), and 20-day Bollinger Bands (mapping dynamic volatility channels).

## 4.2 Environment Modeling

The trading problem is formulated as a Markov Decision Process (MDP). The state space incorporates the current account balance, shares held, and the mentioned market indicators. To provide the agent with short-term historical context, observations are stacked across 10 timesteps using `VecFrameStack`. The agent operates within a continuous action space bounded between 0.0 (100% cash) and 1.0 (100% stock), which is also reinforced by a hardcoded 3% stop-loss mechanism to automatically manage severe downside risk. The agent is also capable of performing 1 action per day, out of the 3 following actions: Buy, Hold, and Sell.

The reward function utilizes the Logarithmic Alpha:

$$R_t = 1500 \times \left( \ln\left(\frac{NW_t}{NW_{t-1}}\right) - \ln\left(\frac{P_t}{P_{t-1}}\right) \right) \tag{1}$$

where $NW$ is the portfolio's Net Worth and $P$ is the SPY close price. While the Sharpe Ratio is excellent for final evaluation, using it as a step-by-step reward penalizes upside

volatility. The Logarithmic Alpha purely rewards market outperformance without discouraging highly profitable upward spikes, while the 1500 scaling factor ensures stable gradient updates during network training.

## 4.3 Agent Modeling

This study evaluates three distinct Deep Reinforcement Learning algorithms, all of which share an Actor-Critic architecture where an "actor" network dictates the trading policy and a "critic" network evaluates the quality of the chosen actions.

### 4.3.1 Advantage Actor-Critic (A2C)

Advantage Actor-Critic (A2C) serves as the foundational baseline for this study, utilizing a synchronous policy gradient approach. The architecture relies on the advantage function, which measures the difference between the actual return and the critic's predicted return, to evaluate action quality. By using this advantage metric rather than raw sequential returns, A2C reduces the high variance inherent in financial market data. This algorithm was chosen to provide a computationally efficient and robust benchmark against which more complex models could be measured.

### 4.3.2 Deep Deterministic Policy Gradient (DDPG)

Deep Deterministic Policy Gradient (DDPG) is an off-policy algorithm explicitly designed for continuous action spaces, perfectly aligning with our continuous fractional portfolio allocation framework (0.0 to 1.0). A key structural advantage of DDPG is its reliance on a replay buffer to store past trading transitions. By sampling random mini-batches from this memory bank during training, the algorithm breaks temporal correlations in the sequential market data, which prevents the agent from overfitting to localized market conditions or recent short-term trends.

### 4.3.3 Proximal Policy Optimization (PPO)

Proximal Policy Optimization (PPO) is an advanced policy gradient method implemented to address the unstable learning rates frequently encountered in highly stochastic financial environments. PPO incorporates a specialized clipping function into its objective to mathematically constrain the magnitude of network updates. This ensures that the newly updated trading strategy does not deviate drastically from the previous iteration. PPO was selected for its proven ability to prevent the model from forgetting amid crashes and guarantee stable convergence despite significant market noise.

# 5 Results and Discussion

The primary evaluation metric is the Sharpe Ratio, which measures risk-adjusted returns:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

where $R_p$ is the portfolio return, $R_f$ is the risk-free rate, and $\sigma_p$ is the standard deviation of the portfolio's excess return. Agent performance was benchmarked against a passive

Buy & Hold strategy, assuming a $10,000 initial investment in the S&P 500 (SPY) held entirely throughout the 2025 out-of-sample period.

Table 1: Final Performance Metrics (Out-of-Sample, 2025)

| **Strategy** | **Final Net Worth ($)** | **Return (%)** | **Sharpe Ratio** |
|---|---|---|---|
| Buy & Hold (Baseline) | 11,855.65 | +18.56% | 1.00 |
| A2C Agent | 12,444.20 | +24.44% | 1.31 |
| DDPG Agent | 12,059.16 | +20.59% | 1.32 |
| **PPO Agent** | **12,506.12** | **+25.06%** | **1.37** |

As detailed in Table 1, all three DRL agents successfully outperformed the baseline strategy in both absolute profit and risk-adjusted returns. While the baseline strategy yielded an 18.56% return with a Sharpe Ratio of 1.00, the A2C, DDPG, and PPO agents achieved Sharpe Ratios of 1.31, 1.32, and 1.37, respectively.

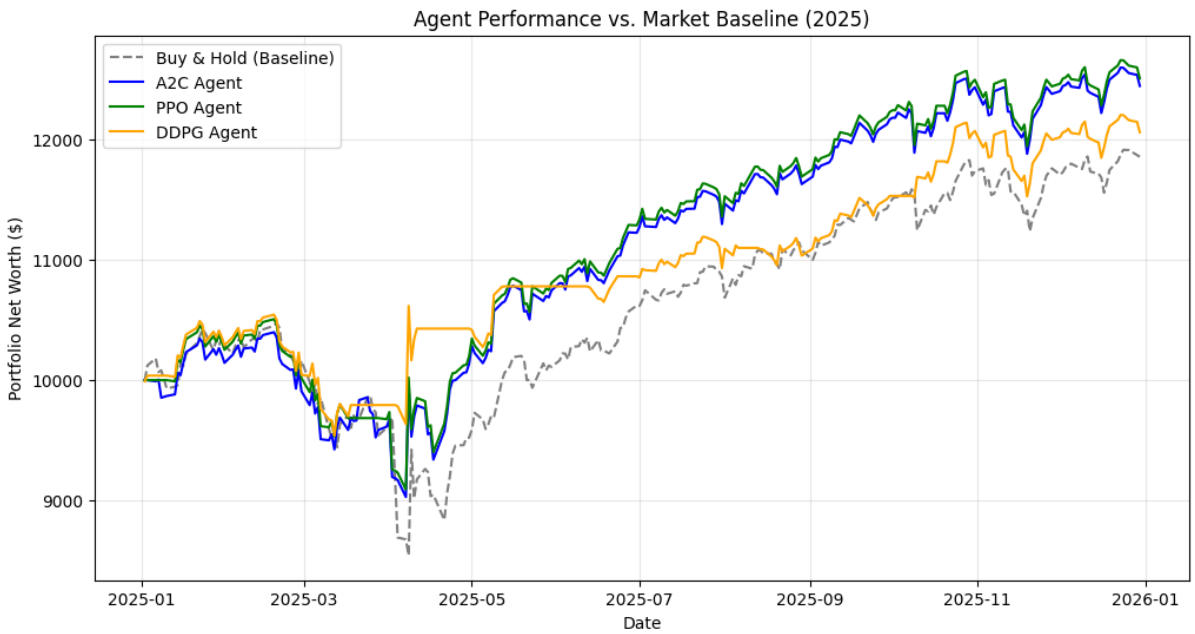


Figure 1: Portfolio Net Worth evolution of the DRL agents vs. Baseline in 2025.

The net worth trajectory depicted in Figure 1 visually demonstrates the behavioral differences between the agents and the baseline. During the significant market drawdown observed around April 2025, the passive baseline strategy suffered a severe decline. Conversely, the DRL agents successfully mitigated this downside risk. Guided by their momentum indicators and the structural 3% stop-loss mechanism, the agents reduced market exposure, which flattened their net worth curves during the crash.

Following the April drawdown, the A2C and PPO agents rapidly capitalized on the market recovery, compounding their capital from a higher preserved base. Ultimately, the PPO agent demonstrated the highest robustness and profitability. Its conservative policy update mechanism involving the clipping function allowed it to efficiently navigate market volatility without drastically overreacting to short-term noise, which resulted in the highest final net worth and optimal risk-adjusted performance.

# 6 Conclusion and Future Work

This project successfully demonstrates the viability of Deep Reinforcement Learning (DRL) for automated financial trading. By formulating the custom trading environment with its own Logarithmic Alpha reward function, all three DRL agents learned to consistently outperform the traditional Buy & Hold baseline in both absolute profit and risk-adjusted returns. Notably, Proximal Policy Optimization (PPO) emerged as the most robust architecture for navigating the stochastic nature of the S&P 500.

While these results are highly promising, several avenues remain for future research. Logical next steps include expanding the agent's action space to incorporate short selling, which would enable profitability during prolonged market downturns. Furthermore, transitioning from daily data to minute-by-minute intraday data would facilitate higher-frequency decision-making, allowing the models to capitalize on micro-momentum trends and execute multiple optimal trading actions per day.

# Source Code

Project code can be found under the following github link:
https://github.com/andrewbejjani/RLTradingAgent

# References


[1] A. Belantari. Deep reinforcement learning-ppo-portfolio optimization, 2025.

[2] IEEE ICECCME. Actor-critic methods in stock trading: A comparative study. In *2023 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*. IEEE, 2023.

[3] Hongyang Yang, Xiao-Yang Liu, Shan Zhong, and Anwar Walid. Deep reinforcement learning for automated stock trading: An ensemble strategy. *arXiv preprint arXiv:2511.12120*, 2025.